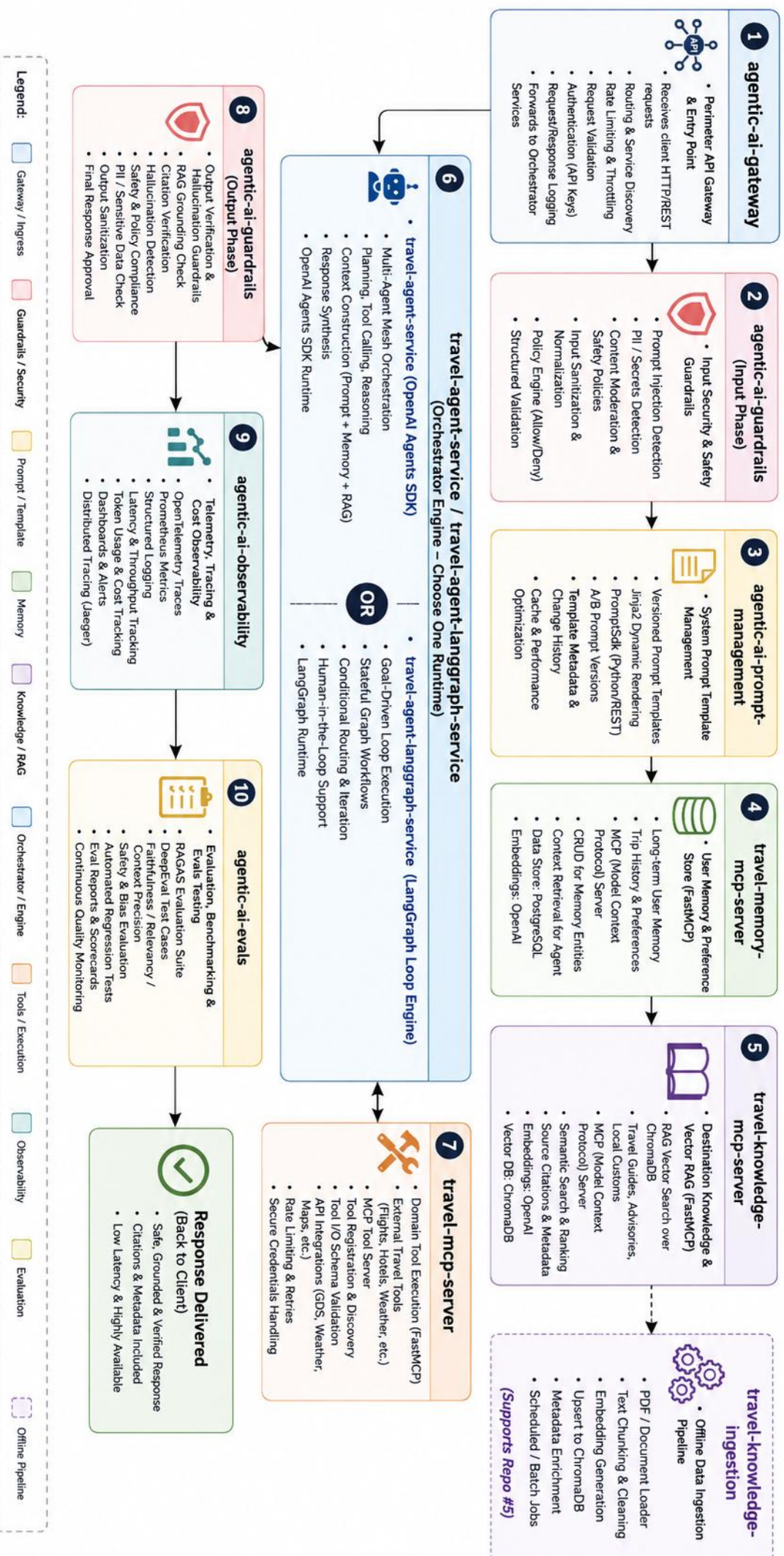


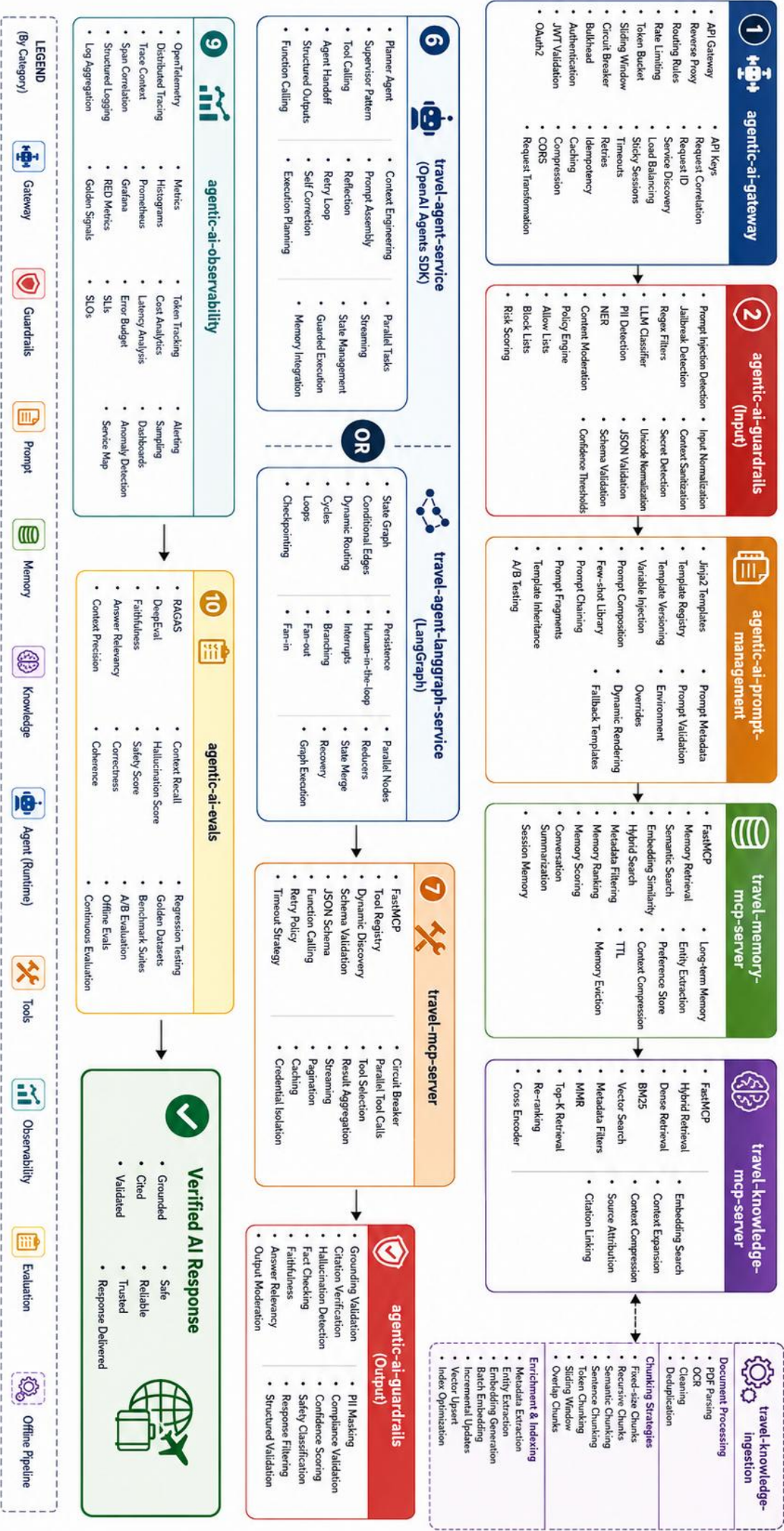
Travel Planner v2 – End-to-End Request Execution Flow (All Repositories)

Sequential flow from client request ingress to response delivery and continuous evaluation



TRAVEL PLANNER V2 – TECHNICAL STRATEGIES ACROSS REPOSITORIES

End-to-End Request Execution Flow (Technical Strategies at Each Stage)



RAG (Retrieval-Augmented Generation) – End-to-End Request Flow & Technical Strategies

Comprehensive RAG Ecosystem: Concepts, Pipeline, Strategies, Frameworks & Evaluation

RAG Evolution / Paradigms



Embedding Models

- OpenAI Embeddings
- BGE
- E5
- GTE
- Jina Embeddings
- Nomic Embeddings
- Cohere Embeddings
- Voyage AI Embeddings
- Instructor Models
- Multilingual Models

Frameworks

- LangChain
- LlamaIndex
- Haystack
- OpenAI Agents SDK
- LangGraph
- DSPy
- CrewAI
- AutoGen
- Semantic Kernel

Parsing Libraries

- Unstructured
- LlamaParse
- Docling
- Apache Tika
- PyMuPDF
- pdfplumber
- Marker
- BeautifulSoup

Observability

- Langfuse
- LangSmith
- Phoenix
- Weights & Biases
- Arize AI
- Monitoring & Alerts
- Tracing & Logging

1 Data Sources

- PDFs
- Word Documents
- Excel
- PowerPoint
- HTML
- Markdown
- Emails
- Wikis
- Confluence
- SharePoint
- Notion
- GitHub
- Databases
- APIs
- CSV
- JSON
- XML
- Images / Audio / Video

2 Data Ingestion

- ETL / ELT
- Batch Ingestion
- Streaming Ingestion
- Incremental Ingestion
- CDC (Change Data Capture)
- OCR
- Layout Parsing
- Table Extraction
- Image Captioning
- Document Parsing
- Data Validation
- Metadata Extraction
- Content Filtering
- Deduplication
- Scheduling
- Error Handling
- Monitoring
- Audit Logging

3 Parsing

- Unstructured
- LlamaParse
- Docling
- Apache Tika
- PyMuPDF
- pdfplumber
- BeautifulSoup
- Layout Analysis
- Table Parsing
- Image Extraction
- Code Parsing (AST)
- Text Extraction
- Section Detection
- Header / Footer Detection
- Language Detection
- Encoding Detection

4 Cleaning & Normalization

- Text Cleaning
- Noise Removal
- Stopword Removal
- Spell Correction
- Lowercasing
- Stemming / Lemmatization
- Language Normalization
- Formal Normalization
- Whitespace Handling
- Deduplication
- PI Reduction
- HTML Removal
- Unicode Normalization
- Encoding Normalization
- Data Standardization
- Entity Standardization
- Disacide Detection

5 Metadata Enrichment

- Metadata Extraction
- Entity Extraction
- Keyword Extraction
- Topic Modeling
- Summary Generation
- Classification
- Tagging
- Custom Metadata
- Source Attribution
- Document Type
- Author / Owner
- Created / Modified
- Security Labels
- Language
- Lineage Tracking
- Confidence Scores
- Business Glossary Mapping

6 Chunking

- Fixed-size Chunking
- Recursive Chunking
- Semantic Chunking
- Sentence Chunking
- Paragraph Chunking
- Section Chunking
- Sliding Window
- Overlapping Chunks
- Parent-Child Chunking
- Hierarchical Chunking
- Token-based Chunking
- Metadata Chunking
- AST Chunking (Code)
- Dynamic Chunking
- Content-aware Chunking
- Adaptive Chunking
- Chunk Merging
- Smart-to-Big Chunking

Embeddings

- Dense Embeddings
- Sparse Embeddings
- Hybrid Embeddings
- Embedding Models
- Embedding Dimensions
- Cosine Similarity
- Dot Product
- Euclidean Distance
- Dot Product
- Dimensionality Reduction
- Multilingual Embeddings
- Domain-specific Embeddings
- Batch Embedding
- Incremental Embedding
- Embedding Cache
- Dimensionality Reduction

8 Indexing

- Vector Index
- Inverted Index
- HNSW
- IVF
- IVF-PQ
- Flat Index
- FAISS Index
- Annoy
- Scann
- DiskANN
- Product Quantization (PQ)
- Scalar Quantization (SQ)
- ANN (Approx. NN)
- ENN (Exact NN)
- Index Optimization
- Index Compression

9 Vector Database / Storage

- ChromaDB
- Pinecone
- Weaviate
- Milvus
- Qdrant
- pgvector
- Elasticsearch
- OpenSearch
- Redis Vector
- Vespa
- LanceDB
- Neo4j Vector
- Azure AI Search
- Multi-store Setup
- ElasticSearch
- Hot / Cold Storage
- Backup & Restore

10 Storage Strategies

- Metadata Filtering
- Namespaces
- Collections
- Sharding
- Partitioning
- Replication
- Encryption
- Multi-tenancy
- Versioning
- Snapshooting
- Compression
- Quantization
- TTL (Time-to-Live)
- Lifecycle Policies
- Access Control
- Data Residency
- Backup Strategy
- Disaster Recovery

11 Retrieval

- Dense Retrieval
- Sparse Retrieval
- Hybrid Retrieval
- Keyword Retrieval
- BM25
- ANN Search
- Metadata Filtering
- Boolean Filtering
- Parent Document Retrieval
- Multi-query Retrieval
- Multi-vector Retrieval
- Self-query Retrieval
- Query Routing
- Query Expansion
- Query Rewriting
- HyDE
- Recursive Retrieval
- Graph Retrieval

12 Re-ranking

- Cross-Encoder
- Bi-Encoder
- MMR
- Reciprocal Rank Fusion (RRF)
- Learning-to-Rank (LTR)
- Rank-based Re-ranking
- Business Rules Boost
- Hybrid Re-ranking
- LLM Re-ranking
- Score Normalization
- Diversity Re-ranking
- Contextual Re-ranking
- Freshness Boosting
- Business Rules Boost

13 Context Optimization

- Context Compression
- Context Packing
- Context Window
- Optimization
- Mitigation
- Hierarchical Context
- Token Budgeting
- Dynamic Context Selection
- Relevance Trimming
- Summary-based Compression
- Token Budgeting

14 Prompt Augmentation

- Prompt Templates
- Retrieved Context Injection
- Citation Prompting
- Few-shot Prompting
- Chain-of-Thought
- Structured Prompting
- Instruction Tuning Prompts
- Role-based Prompts

15 LLM Generation

- Grounded Generation
- Structured Output
- JSON Mode
- Function Calling
- Tool Calling
- Streaming Generation
- Temperature Control
- Top-p / Top-k
- Stop Sequences

16 Guardrails

- Grounding Validation
- Hallucination Detection
- Reference Check
- Faithfulness
- Toxicity Check
- Safety Classification
- PII Detection
- Prompt Injection Detection
- Jailbreak Detection
- Output Validation

17 Grounding & Citations

- Grounding Check
- Document Grounding
- Citation Generation
- Citation Linking
- Span-level Grounding
- Source Attribution
- Reference Tracking
- Citation Deduplication
- Quote Extraction
- Evidence Highlighting
- Confidence Scoring
- Citation Formatting

18 Evaluation

- RAGAS
- DeepEval
- Trulens
- LangSmith Eval
- Precision
- Recall
- Context Precision
- Context Recall
- nDCG
- Hit Rate
- Answer Relevancy
- Correctness
- Coherence
- Hallucination Rate
- User Satisfaction
- Continuous Eval

Feedback Loop

- User Feedback
- Re-train / Fine-tune
- Improve Retrieval
- Update Index
- Active Learning
- Continuous Eval

Optimization Strategies

- Prompt Caching
- Embedding Cache
- Result Cache
- Response Cache
- Incremental Indexing
- Lazy Indexing
- Batch Embedding
- Multi-batch Retrieval
- Parallel Retrieval
- Asynchronous Pipelines
- Query Caching
- Adaptive Retrieval
- Load Balancing
- Cost Optimization
- Token Optimization

Advanced Concepts



Legend (Pipeline Stages)

- 1 Data Sources
- 2 Ingestion
- 3 Parsing
- 4 Cleaning
- 5 Metadata
- 6 Chunking
- 7 Embeddings
- 8 Indexing
- 9 Storage
- 10 Storage Strategies
- 11 Retrieval
- 12 Re-ranking
- 13 Context Optimization
- 14 Prompt Augmentation
- 15 Generation
- 16 Guardrails
- 17 Grounding
- 18 Evaluation
- 19 Optimization

★ This diagram represents a comprehensive enterprise RAG ecosystem with end-to-end pipeline, strategies, tools, and evaluation.

MCP (Model Context Protocol) – End-to-End Request Flow & Technical Strategies

Comprehensive MCP Ecosystem: Architecture, Lifecycle, Capabilities, Tools, Resources, Security, Observability & Integration



Architecture Patterns

- Client-Server Architecture
- Host Architecture
- Tool / Resource / Prompt Providers
- Multi-Server Architecture
- Local / Remote / Distributed MCP
- Gateway Pattern
- Proxy / Pattern

Transport Options

- STDIO
- HTTP / HTTPS
- Server-Sent Events (SSE)
- Streamable HTTP
- WebSocket
- Unix Socket
- Named Pipes
- TCP / IP

Capabilities

- Tool Capabilities
- Resource Capabilities
- Prompt Capabilities
- Sampling
- Logging
- Progress Reporting
- Cancellation
- Pagination
- Notifications
- Completion

Security

- Authentication / Authorization
- OAuth / API Keys / Tokens
- Access Control / RBAC
- Least Privilege
- Input / Output Validation
- Sandboxing
- Secrets Management
- Permission Scopes

Discovery & Connection

- Server Discovery
- Service Registry
- Connection Establishment
- Transport Selection
- Health Check

Strategies

- DNS / Service Discovery
- MCP Registry
- Static Configuration
- Heartbeat / Keep Alive
- Connection Pooling

Initialization & Handshake

- Client Initialization
- Server Startup
- Protocol Version Check
- Capability Negotiation
- Authentication Start

Strategies

- JSON-RPC 2.0 Handshake
- Version Negotiation
- Capability Exchange
- Client / Server Info
- Initialization Options

Authentication & Authorization

- Authentication
- Authorization
- Token Validation
- Access Control
- Session Establishment

Strategies

- OAuth 2.1 / OpenID
- API Keys / Tokens
- RBAC / ABAC
- Permission Scopes
- Secret Management

Discovery & Registration

- Tool Discovery
- Resource Discovery
- Prompt Discovery
- Registration
- Dynamic Updates

Strategies

- Tool Registration
- Resource Registration
- Prompt Registration
- Dynamic Discovery
- Change Notifications

Communication & Messaging

- JSON-RPC Messaging
- Request / Response
- Notifications
- Streaming
- Bidirectional Channel

Strategies

- JSON-RPC 2.0
- Streaming (SSE / WS)
- Message Routing
- Session Management
- Correlation IDs

Capabilities & Operations

- Tool Invocation
- Resource Access
- Prompt Resolution
- Sampling Requests
- Progress Reporting

Strategies

- Tool Calling
- Resource URIs / Templates
- Prompt Templates
- Sampling (Model Calls)
- Pagination / Completion

Context Management

- Context Sharing
- Context Injection
- Context Synchronization
- Context Window Mgmt
- Context Isolation

Strategies

- Context Propagation
- Context Compression
- Context Expansion
- Context Isolation
- Cross-Agent Sharing

Sampling (Model Execution)

- Model Sampling
- Parameter Control
- Sampling Request
- Sampling Response
- Result Streaming

Strategies

- Temperature / Top-p / Top-k
- Max Tokens / Stop Sequences
- Sampling Request / Response
- Streaming Responses
- Parallel Sampling

Streaming & Notifications

- Server Notifications
- Progress Updates
- Real-time Streaming
- Event Delivery
- Backpressure Handling

Strategies

- Server-Sent Events (SSE)
- WebSocket Streaming
- Progress Tokens
- Backpressure Control
- Event Filtering

Error Handling & Resilience

- JSON-RPC Errors
- Validation Errors
- Transport Errors
- Timeout Handling
- Retry / Recovery

Strategies

- Error Codes
- Retry Logic
- Circuit Breaker
- Graceful Degradation
- Exception Handling

Resource Management & Reliability

- Rate Limiting
- Quotas
- Timeouts
- Retries
- Resource Cleanup

Strategies

- Rate Limiting / Throttling
- Quotas / Limits
- Timeouts / Deadlines
- Retry Policies
- Cleanup / Finalizers

Observability & Monitoring

- Logging
- Tracing
- Metrics
- Health Checks
- Audit Logging

Strategies

- Structured Logging
- Distributed Tracing
- Metrics / Dashboards
- Health Checks
- OpenTelemetry

Shutdown & Cleanup

- Graceful Shutdown
- Session Termination
- Resource Release
- Connection Close
- Cleanup Tasks

Strategies

- Graceful Termination
- Session Cleanup
- Resource Cleanup
- Connection Close
- Final Notifications

Tools

- Tool Registration
- Tool Discovery
- Tool Invocation
- Tool Arguments
- Tool Validation
- Structured Output
- Error Handling
- Tool Chaining
- Dynamic Tools

Resources

- Resource URI
- Resource Templates
- Static / Dynamic Resources
- Binary / Text Resources
- Resource Metadata
- Resource Versioning
- Resource Subscription

Prompts

- Prompt Templates
- Prompt Variables
- Dynamic Prompts
- Prompt Discovery
- Prompt Composition
- Prompt Registry
- Prompt Parameters

Context

- Context Sharing
- Context Synchronization
- Context Window
- Context Injection
- Content Propagation
- Context Isolation
- Context Compression
- Context Expansion

Sampling

- Model Sampling
- Temperature
- Top-p / Top-k
- Max Tokens
- Stop Sequences
- Sampling Request
- Sampling Response

Development

- MCP SDKs (Python/JS/Java)
- Inspector / Debugger
- Testing / Mock Servers
- Schema Validation
- Code Generation
- Debugging / Tracing

Performance

- Caching
- Connection Reuse
- Streaming Responses
- Lazy Loading
- Parallel Tool Calls
- Async Processing
- Load Balancing
- Batching

Best Practices

- Stateless Servers
- Idempotent Tools
- Fine-Grained Tools
- Secure Defaults
- Versioning
- Schema Validation
- Backward Compatibility
- Retry Policies / Timeouts

Advanced Concepts

- Dynamic Discovery
- Tool Composition
- Nested MCP Servers
- Federated MCP
- Event-Driven MCP
- Streaming Tool Calls
- Long-Running Tasks
- Human Approval Workflows
- Distributed Context Sharing
- Protocol Extensions

Integration Ecosystem

- OpenAI Agents SDK
- Claude Desktop
- Cursor
- VS Code
- Windsurf
- Continue.dev
- LangChain
- LangGraph
- LlamaIndex
- Semantic Kernel
- CrewAI
- AutoGen

Server Types (Examples)

- Filesystem Server
- Database Server
- PostgreSQL Server
- Git Server
- Rollie Server
- Slack Server
- Calendar Server
- Jira Server
- Confluence Server

Data Access

- File System Access
- Database Drivers
- REST APIs
- GraphQL
- Cloud APIs
- Cloud Storage
- Cloud RPIs
- Vector Database Access
- Object Databases

AI Agent Integration

- Multi-tenancy
- Auditability
- RBAC
- Governance
- Service Mesh
- AI Gateway
- Zero Trust

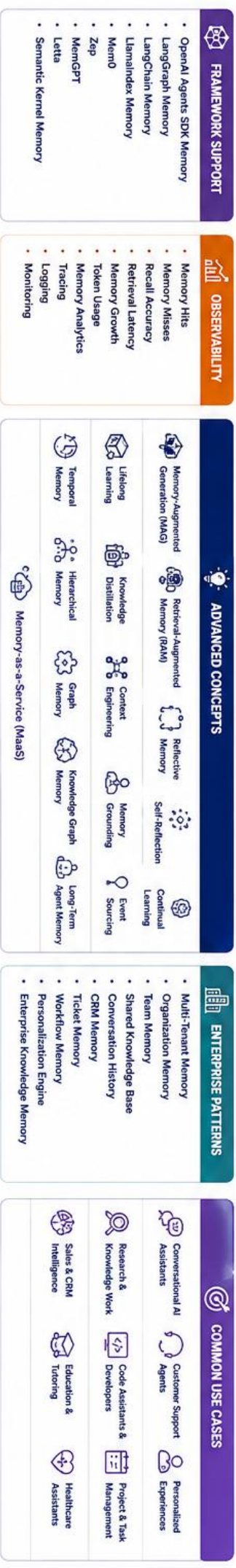
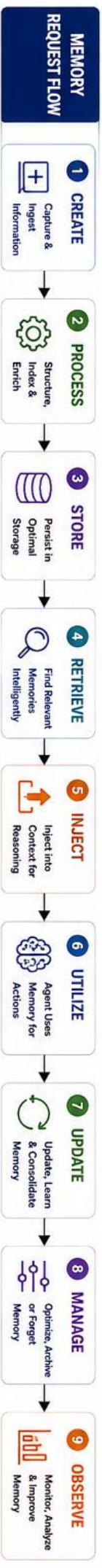
Legend (Pipeline Stages)

- 1 Discovery & Connection
- 2 Initialization & Handshake
- 3 Authentication & Authorization
- 4 Discovery & Registration
- 5 Communication & Messaging
- 6 Capabilities & Operations
- 7 Context Management & Sharing
- 8 Sampling (Model Execution)
- 9 Streaming & Notifications
- 10 Resource Management & Reliability
- 11 Shutdown & Cleanup

MCP enables secure, scalable, and interoperable connectivity between AI models and external tools, data, and prompts through a standardized protocol.

Agentic AI Memory Architecture – End-to-End Memory Lifecycle & Technical Strategies

Comprehensive Memory Ecosystem: Types, Storage, Retrieval, Strategies, Optimization, Security, Observability & Enterprise Patterns



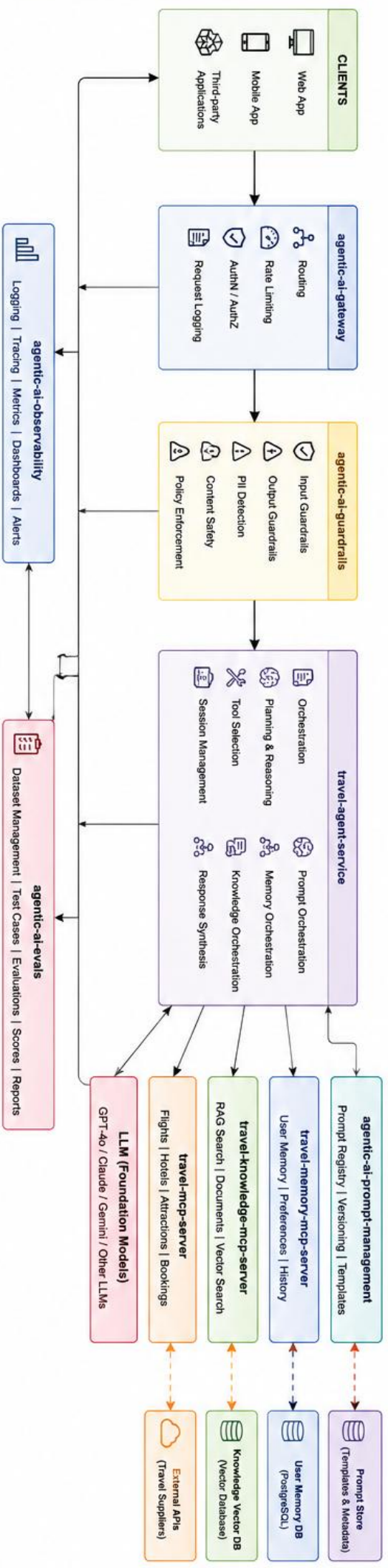
Legend (Memory Lifecycle Stages): 1 Create 2 Process 3 Store 4 Retrieve 5 Inject 6 Utilize 7 Update 8 Manage 9 Observe

★ Effective memory empowers agents with continuity, personalization, and intelligence.

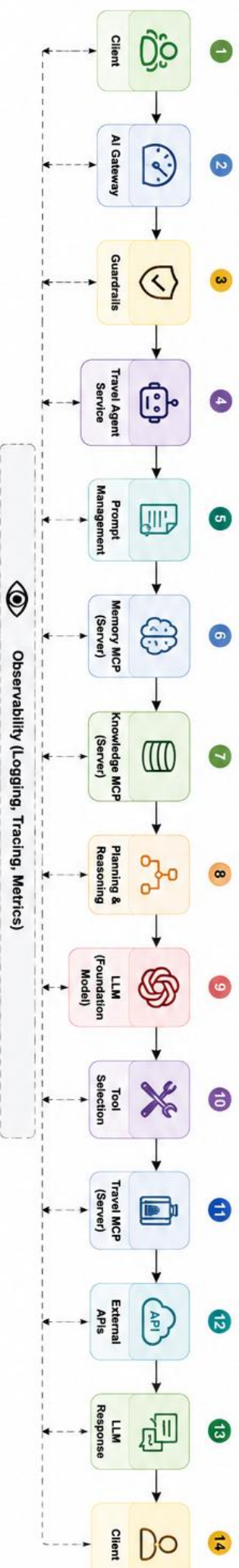
Comprehensive Orchestration Ecosystem: Planning, Execution, State, Multi-Agent Coordination, Tools, Memory, RAG, MCP, Observability & Enterprise Patterns



1. ENTERPRISE AI PLATFORM ARCHITECTURE



2. END-TO-END REQUEST FLOW (RUNTIME)



3. LAYERED ENTERPRISE AI ARCHITECTURE



LLM (Large Language Model) Concepts – End-to-End Architecture & Engineering Reference

From Input to Intelligent Output: Core Concepts, Training, Inference, Interference, Optimization, Evaluation & Operations

1. MODEL ARCHITECTURES

Model Families	Popular Models	Model Components
• Decoder-Only (GPT, Llama) • Encoder-Only (BERT, ES) • Encoder-Decoder (T5, FLAN) • Mixture of Experts (MoE) • Hybrid Architectures	• GPT-4 / GPT-4o • Llama 3.x (8B, 70B, 405B) • Mistral / Mistral • Qwen 2 / Qwen 2.5 • Gemini 2 • Claude 3 (Haiku, Sonnet, Opus) • Phi-3 / Phi-3.5 • DBRX / Grok / Others	• Embedding Layer • Transformer Blocks (N) • Attention Heads (H) • Feed Forward Network (FFN) • Layer Normalization (LN) • Output Projection (LM Head)

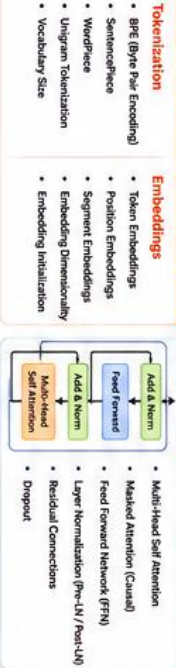
2. TRAINING PIPELINE

Pre-Training	Alignment	Model Compression
• Unsupervised Learning • Next Token Prediction • Large Scale Corpora • Data Deduplication • Tokenization at Scale	• RLHF (PRO) • DPO (Direct Preference Opt) • Constitutional AI • Reward Modeling • Safety Fine-Tuning	• Knowledge Distillation • LoRA / QLoRA • Adapters / Prefix Tuning • PEFT (Parameter Efficient Fine-Tuning)

3. CONTEXT & POSITIONALING

- Context Window
- Maximum Sequence Length
- RoPE (Rotary Positional Emb.)
- ALiB (Attention with Linear Biases)
- Absolute Positional Encoding
- Sliding Window Attention
- Chunked / Block Attention
- RoPE (Rotary Positional Emb.)
- Long Context Extension
- Context Packing
- Context Window Management

4. TOKENIZATION & EMBEDDINGS



5. ATTENTION MECHANISM

- Scaled Dot-Product Attention
- Multi-Head Attention (MHA)
- Query, Key, Value (Q, K, V)
- Attention Weights
- Softmax
- Causal Masking
- Attention Bias / Masks
- Grouped Query Attention (GQA)
- Multi-Query Attention (MQA)
- FlashAttention / FlashAttention-2

6. FEED FORWARD NETWORK

- Position-wise FFN
- Activation Functions (GELU, SiLU)
- Hidden Dimension (d_{ff})
- SwiGLU / GEGLU
- Mixture of Experts (MoE)

END-TO-END LLM INFERENCE FLOW



8. INFERENCE OPTIMIZATION

- KV Cache (Key-Value Cache)
- Prefix Caching
- Speculative Decoding
- Early Exit / Layer Skipping
- FlashAttention / FlashDecoding
- Quantization (INT8 / INT4)
- CPU / RAM / N/Ave Offload
- Continuous Batching
- Paged Attention
- Tensor Parallelism
- Pipeline Parallelism
- Quantization (INT8 / INT4)
- CPU / RAM / N/Ave Offload

9. SAMPLING & DECODING

- Greedy Search
- Temperature Sampling
- Top-k Sampling
- Top-p (Nucleus) Sampling
- Min-p Sampling
- Length Penalty
- Beam Search
- Repetition Penalty
- Presence Penalty
- Frequency Penalty
- Stop Sequences
- Length Penalty

10. SERVING & RUNTIME

- vLLM
- Tensortrt-LLM
- SGLang
- Ollama
- LMDeploy
- Ray Serve
- Custom Inference Server
- Hugging Face Inference
- OpenAI API
- Azure OpenAI
- Groq / Together AI
- Prompt Engineering
- Few-shot / Zero-shot
- System Prompt Design
- Prompt Templates

11. CONTEXT ENGINEERING

- Context Window Management
- Prompt Engineering
- Few-shot / Zero-shot
- System Prompt Design
- Prompt Templates
- Context Compression
- RAG / Memory Injection
- Chunking / Packing
- Context Trimming
- Salience / Relevance

12. OBSERVABILITY & MONITORING

- Time to First Token (TTFT)
- Tokens Per Second (TPS)
- Time Per Output Token (TPOUT)
- Latency (P50 / P95 / P99)
- Throughput
- Token Usage
- Cost Monitoring
- Error Rate / Failure Rate
- Model Performance
- Tracing (OpenTelemetry)
- Request / Correlation ID
- Execution Logs
- Chat / Alignment Benchmarks
- MT-Bench, AlpacaEval, Arena, LMSYS
- Safety & Robustness
- Red Teaming, Jailbreak Benchmarks, Bias, Toxicity, Fairness

13. EVALUATION

- General Benchmarks
- MMLU, HellSwag, ARC, TruthfulQA, GSM8K, HumanEval, BBH, GPQA
- Chat / Alignment Benchmarks
- MT-Bench, AlpacaEval, Arena, LMSYS
- Safety & Robustness
- Red Teaming, Jailbreak Benchmarks, Bias, Toxicity, Fairness

14. SAFETY & ALIGNMENT

- Content Moderation
- Prompt Injection Detection
- Jailbreak Detection
- PII Detection / Redaction
- Toxicity / Hate / Bias Detection
- Guardrails
- Constitutional AI
- Policy Enforcement
- Human Feedback Loop
- Safety Fine-Tuning

15. ERROR HANDLING & RESILIENCE

- Retry & Exponential Backoff
- Timeout Handling
- Circuit Breaker
- Fallback Models
- Graceful Degradation
- Dead Letter Queue
- Exception Routing
- Compensation / Rollback
- Idempotency
- Audit & Recovery

16. MEMORY & STATE INTEGRATION

- Working Memory
- Session Memory
- Persistent Memory
- Short-term / Long-term Memory
- Memory Retrieval
- Memory Updates
- Context Injection
- State Synchronization
- Checkpointing
- State Persistence

17. TOOL & FUNCTION INTEGRATION

- Function Calling / Tool Calling
- Tool Selection
- Tool Routing
- Parallel Tool Calls
- Tool Composition
- Tool Chaining
- Dynamic Tool Discovery
- JSON Schema / Arguments
- Tool Result Handling
- Tool Fallback

18. FRAMEWORKS & ECOSYSTEM

- LangChain
- LlamaIndex
- OpenAI Agents SDK
- CrewAI
- AutoGen
- Semantic Kernel
- Haystack
- RAGFlow
- Others

19. ENTERPRISE & ADVANCED

- Workflow Orchestration
- Human-in-the-Loop
- SLA / SLO Management
- Multi-Tenant Isolation
- Governance & Compliance
- Cost Optimization
- Rate Limiting / Throttling
- Data Privacy (GDPR, SOC2)
- Model Versioning
- A/B Testing / Canary

OPTIMIZATION TECHNIQUES

- Quantization (INT8 / INT4)
- Pruning
- Knowledge Distillation

TRAINING DATA & QUALITY

- Data Collection
- Data Deduplication
- Filtering
- Curriculum Learning
- Data Mining
- Quality Scoring
- Scoring
- Mitigation

DEPLOYMENT OPTIONS

- Cloud (AWS, GCP, Azure)
- On-Premise
- Hybrid
- Edge
- Private Cloud
- Air-Deployed

LEGEND: LLM INFERENCE PIPELINE STAGES

1. Input
2. Tokenization
3. Embeddings
4. Positional Encoding
5. Transformer Layers
6. Attention
7. FFN
8. Normalization
9. Residual Connections
10. Logits
11. Sampling
12. Output Tokens
13. Detokenization
14. Final Response

★ LLMs transform input text into intelligent responses through a complex pipeline of tokenization, neural computation, and generation — powered by data, training, and optimization at scale.